

FactWorld: A Reproducible Instrument for Composing State Tracking and Recall

Ian Barber

July 2026

Abstract

FactWorld is a benchmark suite that measures recall, state tracking, and their composition, identically for frontier models over an API and for small models trained from scratch. Every task must pass two tests: does it differentiate frontier models, and can it be explored architecturally. Every number reproduces from committed scripts and can be checked with an API key or a single GPU. We find that the component abilities are largely solved at the frontier, but their composition is not. With reasoning off, most models hold little composition in their weights and architecture. With reasoning on, composition is bought by the token, and the price differs significantly by model. In task-specific training, each element of the composition requires a specific architectural or training capability.

1 Introduction

Recall and state tracking are usually benchmarked separately. Recall has multi-query associative recall (Arora et al., 2023); state tracking has the S_5 word-problem literature (Barrington, 1989; Liu et al., 2023). Whether a model can compose them, tracking a changing state and then acting through what it tracked, is a different question, and it is the question FactWorld measures.

FactWorld is a frozen, versioned suite of tasks that render to natural language over a constrained vocabulary, with deterministic examples from fixed seeds. Gold answers come from a symbolic oracle applied to the underlying world state, never from parsing rendered text, so labels cannot leak. A validity gate certifies that no shallow shortcut clears floor on any task. One scoring rule and one protocol serve both frontier models over an API and small models trained from scratch, so a finding made in one regime can be checked in the other.

Measured this way, the component abilities are largely solved at the frontier, but their composition is not. With reasoning off, most models hold little composition in their weights and architecture, and the ordering does not follow the reasoning-on ranking. With reasoning on, composition is bought by the token, and the price differs significantly by model. In task-specific training, each element of the composition requires a specific architectural or training capability, and the composed task itself remains unsolved at the scales tested.

The instrument is deliberately narrow. Every task is single-turn and single-answer-span, with no tool use, planning, or multi-turn action, and the results are measurements within the regime tested, not scaling laws.

2 The FactWorld instrument

2.1 Tasks

Each task isolates one component ability or one composition of them (Table 1).

	task	role
Component: recall	<code>recall_copy_v1</code>	deferred-readout MQAR variant; pool breadth is the load axis
Component: recall (parametric)	<code>conflict_v1</code>	memorized map contradicted in context
Component: state tracking	<code>binding_v2</code>	last-write-wins over a give-stream
commutative variant	<code>commutative_v1</code>	per-entity accumulation mod k ; order-free
non-abelian variant	<code>s5_v1</code>	order-sensitive role permutations
Composition: state $\times$ recall	<code>composite_copy_v2</code>	the two-hop; the gap is its derived statistic
Composition: recall $\circ$ recall	<code>chain_v2</code>	pointer chase at fixed breadth, explicit hop count
Composition: state $\times$ dereference	<code>s5_chain_v3</code>	the FactWorldBench headline task

Table 1: The task taxonomy: components first, then their compositions.

Composition (`composite_copy_v2`). A set of facts maps agents to values, and a stream of give events moves objects between agents. The query asks for the value of the agent that currently holds a given object:

```
g2's a0 is v70. g4's a0 is v24. g0's a0 is v109. g1's a0 is v48.
s0 gives o3 to g4. s1 gives o3 to g1. s2 gives o3 to g2.
what is a0 of the holder of o3?
gold: g2 v70 .
```

The model must resolve the holder by tracking the give-stream (last write wins), then look up that holder’s value in the facts. The facts are resampled every example, so the value is read from context, not from weights. The answer is two tokens, a holder and a value, and each leg is scored independently: holder-right/value-wrong means the model tracked state but failed the recall hop, and the reverse means the opposite. This decomposition localizes failures throughout the report.

S_5 (`s5_v1`). A stream of swap and cycle events permutes the roles of a set of agents, and the query asks for one agent’s final role. Swaps and cycles do not commute, so the running permutation must be carried step by step.

`s5_chain` (`s5_chain_v3`). The headline composes the two stress components in one task. $k = 16$ agents hold an `a0` pointer map, initially one 16-cycle. L order-sensitive swap/cycle events permute the pointer values: the S_5 -style state-tracking load. The query then dereferences the final map 8 hops deep (`what is a0 of a0 of ... of gX? (8 hops)`): the pointer-chase load. Neither component alone suffices. Every item is gated so the query start sits on a final-map cycle longer than the query depth: the nine path nodes are distinct, answering the queried agent or any fixed hop scores exactly 0, item difficulty is uniform, and chance is $1/16$. Cycle events are rendered simultaneity-explicit, so no sequential misreading of the assignments is available. Tests pin the gate and the rendering.

Difficulty knobs (length L , pool breadth, chain depth) are calibration parameters, used to place each model class mid-scale.

2.2 Scoring

The metric is **match**: strip a trailing period from both sides and compare the model’s first len(gold) whitespace tokens to the gold answer, binary per item, no partial credit. Containment is the one published diagnostic, and any cell where containment exceeds match by 0.08 or more has its raw predictions read before the number is believed. On reasoning arms, match scores the answer a multi-line emission commits to (a single-token final line, an emphasized answer closing a statement, or a lone token in a trailing code fence), never the working’s first tokens. The commitment is located structurally, never by demanding a rigid output format.

2.3 Regimes and floors

Every frontier model runs in two regimes. **Instant** is reasoning off with a hard one-line answer contract and a 96-token cap: what the weights compute. **Thinking** is a generous reasoning budget at the shared top effort arm: what reasoning buys. The two rankings are near-orthogonal, so profiles are per-axis, never a single scalar.

Instant cells are read against two floors, recomputed at render time from the exact task items. The *recency heuristic* answers the last event’s recipient and scores 0.04. The *object-filter floor* filters events to the queried object but guesses among its writes and scores $\mathbb{E}[1/w]$: 0.41 at L16, decaying roughly as $1/L$. A score near the floor shows object filtering, not state tracking.

The floors exist because samplers leak shortcuts. A give-stream sampler that draws events uniformly leaves the queried object’s resolving write close to the end of the stream, so a one-line recency heuristic scores 0.33, indistinguishable from mid-roster state tracking. The current sampler places the queried object’s last write uniformly over the stream, which drives that heuristic to chance and exposes the object-filter floor as the real bar: cheap-tier models sit at or below 0.41 where they previously looked mid-pack. The validity gate checks the recency heuristic on every give-stream task.

3 Validating the instrument

Before the suite compares architectures or models, it must reproduce the field’s established single-capability dissociations. All three reproduce on the natural-language format, three seeds each.

1-hop associative recall (MQAR). The value is read adjacent to the key. Attention solves it: gdp_hybrid 1.00, fprm 1.00, transformer 1.00 (pool 16).

Deferred readout recall. The value must be read at an arbitrary later position, the regime composition actually requires: gdp_hybrid 0.73, fprm 0.50, transformer 0.19 (pool 5). Every architecture aces 1-hop; only the recurrent hybrid solves deferred readout.

S_5 length extrapolation under dense supervision. Train dense, evaluate free-running past the training length (Table 2). The product recurrence extrapolates; the transformer and the looped block shortcut past the trained length.

4 Benchmarking the frontier

The recurring benchmark reads a thirteen-model roster through the instrument. Instant cells run $n = 100$ under the answer contract. Thinking cells run $n = 25$ with per-length completion budgets

arch	L16 (train)	L64 (4×)	L128 (8×)
gdp_hybrid	1.00	0.90	0.82
fprm	1.00	0.17	0.23
transformer	0.79	0.22	0.22

Table 2: S_5 length extrapolation under dense supervision, three seeds.

sized so truncation, scored as wrong, stays a rounding error; Wilson 95% intervals accompany every cell, and thinking differences under about 0.2 are not an ordering. Models run at the shared top effort arm, **xhigh**, mapped down where the endpoint’s ceiling is **high**. Contamination marks quarantine cells from orderings: $\oslash$ means not measurable at this budget, † means covert or visible reasoning on an instant attempt, ‡ means the provider ignored the token cap. Three models (grok-4.5, muse-spark-1.1, claude-fable-5) are thinking-only: their endpoints cannot disable reasoning, so they carry no instant numbers.

4.1 FactWorldBench

s5_chain_v3 is the ranking task: one number per model for how well it holds a mutating pointer map and then acts through it. The L96 cell ranks the full roster, the L128 cell separates the top cluster, and the matched L64 cell prices completion tokens per call (nearly every model solves L64, so token spend compares like for like).

Model	s5_chain @L96	@L128	ctok/call @L64
anthropic/claude-fable-5	1.00	1.00	5014
openai/gpt-5.5	1.00	1.00	9343
x-ai/grok-4.5	1.00	0.96	7711
anthropic/claude-opus-4.8	0.96	0.96	9702
moonshotai/kimi-k3	0.96	0.96	10941
nvidia/nemotron-3-ultra	0.96 ^r	0.96	17071
muse-spark-1.1	0.96	0.92	12484
google/gemini-3.6-flash	0.92	0.96	8166
anthropic/claude-sonnet-5	0.92	0.96	12729
deepseek/deepseek-v4-pro	0.92	0.96	17052
z-ai/glm-5.2	0.92	0.80	17982
qwen/qwen3.7-max	0.72	0.44	12588
openai/gpt-5.6-sol	0.60	0.80	2444

Table 3: FactWorldBench: **s5_chain_v3**, match, $n = 25$ per cell. ^r marks a single raised-budget rerun.

L96 separates a 0.92 to 1.00 band from a tail. L128 spreads the roster from 0.44 to 1.00: fable and gpt-5.5 hold 1.00, qwen halves to 0.44, glm drops to 0.80. The ^r on nemotron’s L96 is a single raised-budget rerun at 98,304 tokens: at the planned budget it read 0.84 with 12% truncation, budget censoring rather than capability, and at L128 it holds 0.96 clean. Token spend also separates the top cluster: the L64 spend spans 5.0k (fable) to 17.1k (nemotron) per call, a 3.4× range for the same scores.

Serving behavior is part of what a model ships. gpt-5.6-sol’s Chat Completions shim caps the effort

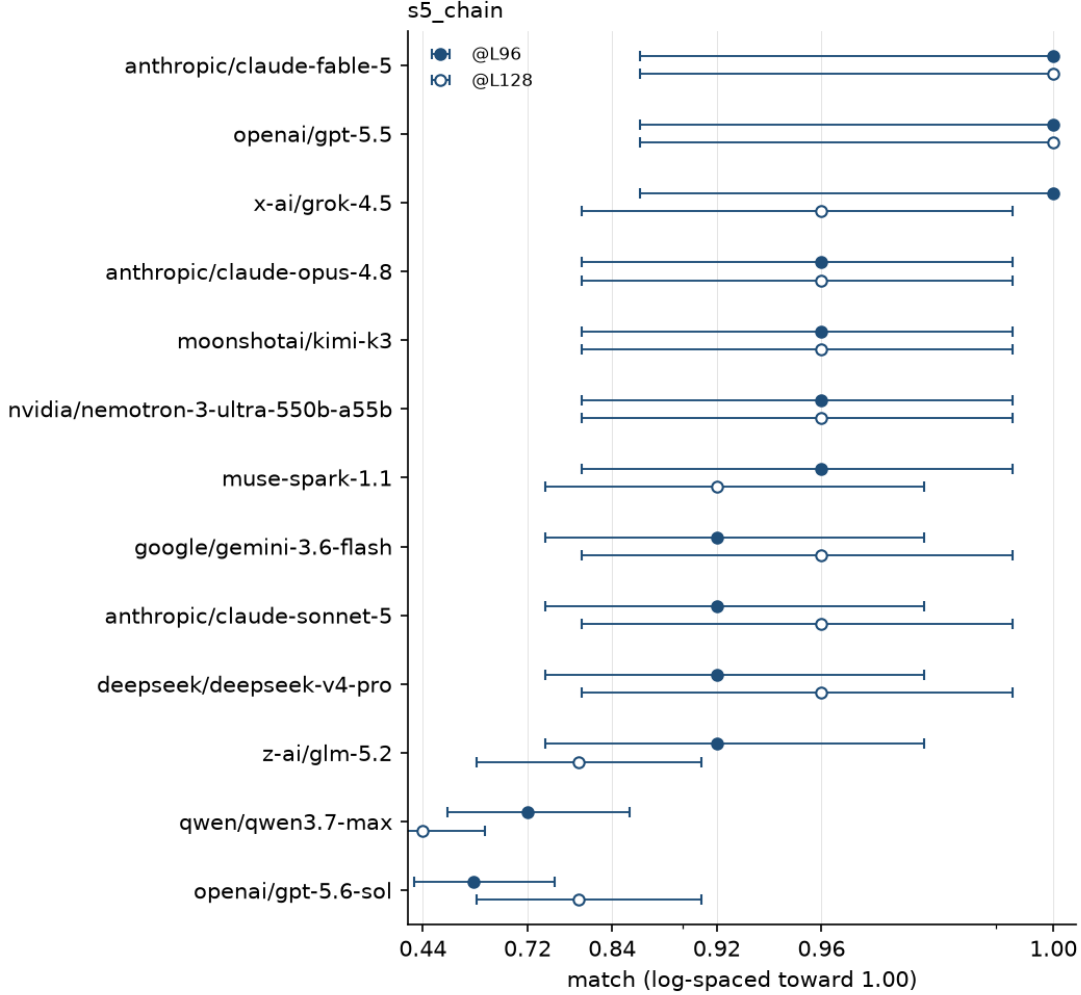


Figure 1: FactWorldBench scores at L96 (filled) and L128 (open) with Wilson 95% intervals, plotted on a log-spaced match axis. Overlapping intervals are not an ordering.

ladder at **xhigh**, while its native Responses API exposes a further **max** level that lifts L96 from 0.60 to 0.88 at $2.3\times$ the reasoning tokens; the scored rows stay on the shared **xhigh** arm for cross-model fairness. Sol’s length curve is also the roster’s only inverted one: it under-allocates thinking to short prompts at every effort level, and those failures are genuine wrong answers (match equals containment).

4.2 The composed cell and the gap (instant regime)

With reasoning off, does the composition exist in weights at all? The composed cell is the two-hop query under the zero-budget protocol at L16 and L64. The **gap**, $\text{binding@L16} - \text{composed@L16}$, is interpretable only where the binding component is established. The recall half is free (0.97 to 1.00 on sanity, load, and scaffolded recall for every measurable model), so the gap is a composition deficit, not a recall one (Table 4).

Where binding is solid the roster separates. Opus composes essentially for free: gap 0.06, equal

Model	recall	binding @L16	composed @L16	composed @L64	gap @L16
anthropic/claude-opus-4.8	1.00	0.78	0.72	0.43	+0.06
anthropic/claude-sonnet-5	0.97	0.77	0.62 [†]	0.32 [†]	+0.15 [†]
deepseek/deepseek-v4-pro	1.00	0.51	0.44	0.19	− ^f
google/gemini-3.6-flash	1.00	0.69*	0.67*	0.26*	+0.02*
moonshotai/kimi-k3	1.00	0.65	0.33	0.29	+0.32
nvidia/nemotron-3-ultra	1.00	0.49	0.33	0.12	− ^f
openai/gpt-5.5	1.00	0.80	0.46	0.33	+0.34
openai/gpt-5.6-sol	1.00	0.82	0.65	0.33	+0.17
qwen/qwen3.7-max	1.00	0.51	0.24	0.08	− ^f
z-ai/glm-5.2	1.00	0.71	0.38 [†]	0.13	+0.33 [†]
<i>recency heuristic (floor)</i>		0.04	0.04	0.06	
<i>object-filter floor</i>		0.41	0.41	0.15	

Table 4: The instant battery, match, $n = 100$. −^f marks a gap that is not interpretable because binding sits at the object-filter floor; * marks an off-arm that ran at effort minimal (the endpoint cannot disable reasoning).

to the instant test-retest noise bar. gpt-5.5 pays the largest clean deficit (binding 0.80, composed 0.46, gap +0.34): the model that shares the top of the thinking headline has among the least in-weights composition. kimi-k3 is the same shape (gap +0.32 against a top-cluster headline). For deepseek, qwen, and nemotron the binding leg’s interval overlaps the 0.41 object-filter floor, so the gap renders −^f: floor minus floor is zero by construction, not a measurement.

4.3 Reasoning buys composition

On the composed cell at L16 ($n = 50$ per cell), reasoning effort gives a clean dose-response (Table 5). At effort none the holder leg reads against the 0.41 object-filter floor: object filtering, not established composition. Low effort already recovers most of the composed cell, and the curve is monotone through high. The lever is implicit reasoning: an explicit “write the holder, then the value” instruction hurts every model, including the reasoners that solve the composed cell under a plain prompt (format-fair ablation, $n = 100$).

model	none	low	medium	high
kimi-k2.6	0.72	1.00	1.00	1.00
glm-5.2	0.38	0.92	0.96	0.98

Table 5: Reasoning-effort dose-response on the composed cell at L16.

4.4 The components under stress

The two state-stress components are read separately in the thinking regime, far past the composed cell’s settings: chain d128 (a 128-hop pointer chase at fixed breadth $k = 257$, chance below 0.01) and S_5 @L256 (256 permutation events, concrete rendering, chance 0.20). The s5@128 ctok column is completion spend on the matched L128 cell every model runs (Table 6).

The top half of the roster holds both components at or near ceiling, which is why the composite, not the components, carries the ranking. The chain query includes an explicit hop count because

Model	chain d128	s5 @L256	s5@128 ctok
x-ai/grok-4.5	n/a	1.00 [‡]	8069
muse-spark-1.1	0.96	1.00 ^r	9704
anthropic/claude-sonnet-5	1.00	1.00 ^r	11866
anthropic/claude-opus-4.8	1.00	1.00 ^r	12683
anthropic/claude-fable-5	1.00	0.96	6405
openai/gpt-5.5	1.00	0.96	6989
openai/gpt-5.6-sol	0.88	0.92	2657
z-ai/glm-5.2	0.92	0.88	6282
google/gemini-3.6-flash	0.96	0.84	8234
moonshotai/kimi-k3	1.00	0.80 ^r	11355
qwen/qwen3.7-max	0.96	0.80	7904
deepseek/deepseek-v4-pro	1.00	⊘	10043
nvidia/nemotron-3-ultra	0.60	⊘	12250

Table 6: Thinking-regime component stress cells, match, $n = 25$. ^r marks a single raised-budget rerun at 32,768 tokens; ⊘ means majority truncation at the standard budget.

the bare nested phrase is a hop-counting confound at depth 128: models miscount the repetitions and stop a few hops short, a prompt-format artifact rather than a state failure.

The components also dissociate the regimes within one item set. The chain d16 cell runs in both regimes on identical items. Thinking, it reads 0.96 to 1.00 for twelve of thirteen models. Instant, every cleanly-answering model floors (best 0.08). A 16-hop pointer chase is serial work no roster model holds in weights, and every strong model solves it given room to work.

4.5 Long context

Both regimes are stressed well past their calibration points. The thinking regime holds composition at long context: glm-5.2 reads 0.94 to 0.98 from L64 out to L1024 on the composed task ($k = 32$, pool 16), while its instant arm sits at or below the object-filter floor from L128 on. At this breadth, length is not the binding constraint for the thinking regime. S_5 under a concrete rendering with reasoning holds 0.90 at L128, degrading gradually (Appendix A).

5 Exploring the architectures

The same tasks train from scratch with next-token prediction. Supervision is answer-only by default, with a staged curriculum for the composite flagship; dense per-step supervision appears only where it is the measured lever (the S_5 and commutative formation results). The architectures:

- **transformer**: a standard decoder-only stack, the attention baseline.
- **gdn_pure** / **gdn_hybrid**: GatedDeltaNet (Yang et al., 2024), gated delta-rule linear attention; pure is attention-free, hybrid interleaves one full-attention layer per four.
- **gdp_pure** / **gdp_hybrid**: GatedDeltaProduct (Siems et al., 2025), the delta rule generalized to a product of Householder transformations per token ($n_h = 4$). This is the non-commutative recurrence the state-tracking results turn on; same pure/hybrid split.

- **fprm**: a weight-tied looped conv+attention block;¹ one block applied repeatedly, so per-token FLOPs match the transformer at 5 to 11× fewer parameters.

Comparisons are matched on compute, not parameters, at budgets sufficient for the capable configuration to converge. At the flagship scale ($d_{\text{model}} = 768$, 8 layers), fprm’s per-token FLOPs equal the transformer’s (~ 159 GFLOP) at 10M vs 76M params, and gdp_hybrid’s Householder-product recurrence costs $1.25\times$ the transformer’s (204 GFLOP at 101M params).

5.1 The composite flagship

On the composed task, pool 16 at L16, with the staged curriculum (25k steps, 80k docs, 3 seeds, eval $n = 500$):

arch	params	per-tok FLOPs	composite @L16	holder leg	value leg
gdp_hybrid	101M	204 GFLOP	0.833 ± 0.089	0.999	0.833
fprm	10M	159 GFLOP	0.109 ± 0.089	0.998	0.109
transformer	76M	159 GFLOP	0.001 ± 0.001	0.065	0.041

Table 7: The composite flagship with per-leg decomposition.

gdp_hybrid solves the binding leg and most of the recall leg, and is competitive with API models on this task despite training from scratch at 100M params. fprm solves binding perfectly and fails to recall the value of the holder it just resolved. The transformer fails both legs. The per-leg decomposition names the deficit: the failure is routing the resolved holder into the in-context recall lookup, the same deficit the frontier’s instant gap measures.

The scaffolded probe sharpens the contrast. Given the correct holder, frontier models recall the value at 0.98 to 1.00; the local flagship reads 0.096. API models can do each leg when the problem is split for them and struggle to compose the legs in one prompt; the local models that fail cannot do the second hop even when handed the first.

5.2 Scale does not fix the composite

The same curriculum at three compute-matched sizes (2 seeds each):

arch	small (384×6)	medium (768×8)	large (1024×12)
gdp_hybrid	0.12 ± 0.08	0.73 ± 0.01	0.21 ± 0.21
fprm	0.12 ± 0.05	0.03 ± 0.01	0.03 ± 0.02
transformer	0.01 ± 0.00	0.01 ± 0.01	0.00 ± 0.00

Table 8: Compute-matched scale sweep on the composed task.

Convergence is architecture-specific and scale-dependent, peaking at medium: gdp_hybrid at 768×8 is the only converging cell. The transformer floors at every scale including 202M params and 417 GFLOP/token, with containment near zero as well: a real floor, not a formatting miss. Wherever binding is solved and the composed cell fails, the value leg is what collapses; the routing deficit is scale-invariant.

¹After Movahedi et al. (2026); we did not run their model.

5.3 Binding under breadth

fprm’s product recurrence leads the binding leg through B16 (1.00 at B6, 0.97 to 0.98 seed-consistent at B16) and stops fitting at B24, where only the gated hybrid holds (0.67); the transformer reads 0.08 to 0.23 throughout (45 runs at $d = 256$). Last-write state requires recurrence, ordered by recurrence form, and breaks by form as the working set grows.

5.4 Chain depth does not extrapolate

Trained at chain depths 2 and 3, no architecture reads above the 1/6 guess at depths 4 and 5 (fprm 0.21/0.15, transformer 0.16/0.09, gdp_hybrid 0.01/0.00; 3 seeds each). gdp_hybrid fits the training distribution best and scores worst held-out: a depth-specific circuit, systematically wrong one hop out, not a guesser. Dense intermediate-hop supervision does not help (0.00 to 0.10 held-out). Over the API the same composition solves at d16, but only in the thinking regime.

5.5 S_5 forms under dense supervision and survives weaning

Under answer-only supervision S_5 floors for every architecture. It forms when the training signal carries the state: interleave the oracle’s state checkpoint every K events and evaluate free-running (10 seeds):

K (stride)	value @L16	value @L64	converge @L16
1 (dense)	1.00	0.75	10/10
2	0.98	0.40	10/10
4	0.19	0.20	0/10
8	0.21	0.20	0/10

Table 9: Supervision density: the circuit forms at a checkpoint every ≤ 2 events and is gone below.

The threshold is sharp. Formation is architecture-independent; length extrapolation is not (§3). The circuit also survives weaning to label-free deployment: train dense, fine-tune on mixed densities including answer-only, and 8/8 seeds converge free-running, extrapolating on par with dense-only. Stressed to $32\times$ the trained length on the dense-supervised non-abelian composite, the recurrent hybrid holds about 0.5 out to L512 while the looped block stays at floor.

5.6 Commutative state

The commutative rung (per-entity accumulation, order-free) does not form under answer-only supervision at any measured setting: chance for every architecture at $d = 256$. Dense per-step traces form it in-distribution for the recurrent architectures (gdp_hybrid 0.82 ± 0.15 , fprm 0.65 ± 0.26 at L16; transformer at chance), and no run carries it past the training lengths. The pattern matches S_5 : supervision density enables formation, and extrapolation is a separate, unmet requirement.

6 What each element of the composition requires

In task-specific training, each element of the composition requires a specific architectural or training capability (Table 10).

element	requirement
adjacent (1-hop) recall	attention: every architecture aces adjacent readout
deferred recall	product recurrence: the transformer aces adjacent, fails deferred (0.19 vs 0.73)
last-write state	recurrence, ordered by form: fprm through B16, only the gated hybrid at B24, transformer floors
non-abelian state (formation)	dense per-step supervision, a checkpoint every ≤ 2 events, architecture-independent
non-abelian state (extrapolation)	recurrent hybrid: gdp_hybrid 0.75 @L64; fprm and transformer collapse past train length
commutative state (formation)	dense per-step supervision plus recurrence; extrapolation does not follow
depth extrapolation	open: unsolved by every measured choice, with or without intermediate-hop traces
local composition (value leg)	open at the default recipe: only the staged curriculum at 768×8 converges it

Table 10: What each element of the composition requires.

Depth extrapolation and the value leg of the composed cell remain unsolved at the scales tested: no measured architectural or training choice provides the first, and only one curriculum at one scale converges the second.

7 Discussion

The same composition probes that rank the frontier roster also separate architectures trained locally, and the per-leg decomposition links the regimes: a finding about routing the resolved holder into recall can be checked in both settings.

The main findings:

- **Composition is where frontier models still separate.** The components are largely solved in the thinking regime, but their composition is not: the headline composite differentiates by score at both ends and by tokens-to-solve within the top cluster. With reasoning off, the composed cell shows most of the roster holds little composition in weights, with an ordering the thinking headline does not predict: gpt-5.5 shares the top of one and pays the largest gap on the other.
- **Composition responds to reasoning, monotonically.** Effort moves the composed cell from near floor to 0.92 and above, and the thinking regime holds it out to L1024. Explicit prompting does not substitute.
- **Non-abelian state tracking responds to reasoning only under a concrete rendering.** Neither reasoning over abstract tokens nor a concrete rendering without reasoning suffices; the combination solves it, with a model-dependent length limit (Appendix A).
- **For local models, S_5 forms under dense supervision.** A state checkpoint at least every two events develops a length-extrapolating circuit that weans to label-free deployment.
- **Architecture carries length generalization.** A learned state circuit generalizes in length only on a recurrent hybrid; transformers and looped blocks shortcut.

These are results within the regime tested, not scaling laws.

8 Limitations and related work

Limitations. The scale regime is bounded: $k = 5$ S_5 , local models of 3M to 269M params matched on compute at 32 to 540 GFLOP/token, pretrained models from a few billion to about a trillion parameters. Composition is 2-hop throughout except the headline’s 8-hop dereference. Instant cells run $n = 100$; thinking cells run $n = 25$ to 50 because API costs scale with reasoning tokens, and thinking differences under about 0.2 are not an ordering. The component-to-agent mapping is a motivating analogy, not a proven one.

Related work. Recall is grounded in multi-query associative recall (Arora et al., 2023). State tracking is grounded in the S_5 word-problem literature: Barrington (1989) makes S_5 -recognition NC¹-complete; Merrill and Sabharwal (2023) and Merrill et al. (2024) bound transformers and SSMs within TC⁰; Liu et al. (2023) show transformers learn depth-limited shortcuts; Grazi et al. (2024) unlock state tracking in linear RNNs via negative eigenvalues; Siems et al. (2025) introduce the Householder-product recurrence our gdp models use; Yang et al. (2024) introduce GatedDeltaNet. FactWorld’s contribution is measuring the components independently and composed, under one protocol, for API models and from-scratch models alike.

9 Reproducibility

Every headline claim maps to a committed script and raw results in the repository. The data, oracle, and eval layer is pure stdlib; training runs need one CUDA GPU. Raw per-cell records (all attempts, usage, diagnostics) are one JSON object per cell in the benchmark history; the rendered feed carries per-cell Wilson intervals, marks, figures, and provenance.

A S_5 at the frontier

Without reasoning, S_5 floors at every length. Models do real step-by-step tracking on individual examples, then stall: they report a role the queried agent held at a recent step rather than the final one. In aggregate this is chance at every length from L4 to L128, for every model measured.

With reasoning on, the outcome depends on the rendering. Reasoning over the abstract token rendering leaves a strong model near 0.33; a concrete rendering (people and jobs) without reasoning is chance; the combination solves the task. Under an 8,192-token budget the concrete-plus-reasoning curve degrades gradually: glm-5.2 holds 1.00 at L32, 0.97 at L64, and 0.90 at L128, with residual errors that are genuine wrong answers, not truncation. The degradation point is model-dependent (kimi-k2.6 degrades sooner), and there is no abrupt break.

Frontier models solve S_5 by reasoning over a concrete rendering. Small models trained from scratch solve it when the training stream carries a state checkpoint at least every two events (§5.5).

References

Arora, S., Eyuboglu, S., Timalsina, A., Johnson, I., Poli, M., Zou, J., Rudra, A., and Ré, C. (2023). Zoology: Measuring and Improving Recall in Efficient Language Models. arXiv:2312.04927.

- Barrington, D.A.M. (1989). Bounded-width polynomial-size branching programs recognize exactly those languages in NC^1 . *Journal of Computer and System Sciences*, 38(1):150–164.
- Grazzi, R., Siems, J., Zela, A., Franke, J.K.H., Hutter, F., and Pontil, M. (2024). Unlocking State-Tracking in Linear RNNs Through Negative Eigenvalues. arXiv:2411.12537.
- Liu, B., Ash, J.T., Goel, S., Krishnamurthy, A., and Zhang, C. (2023). Transformers Learn Shortcuts to Automata. arXiv:2210.10749.
- Merrill, W. and Sabharwal, A. (2023). The Parallelism Tradeoff: Limitations of Log-Precision Transformers. *TACL*, 11:531–545.
- Merrill, W., Petty, J., and Sabharwal, A. (2024). The Illusion of State in State-Space Models. arXiv:2404.08819.
- Siems, J., Carstensen, T., Zela, A., Hutter, F., Pontil, M., and Grazzi, R. (2025). DeltaProduct: Improving State-Tracking in Linear RNNs via Householder Products. arXiv:2502.10297.
- Yang, S., Kautz, J., and Hatamizadeh, A. (2024). Gated Delta Networks: Improving Mamba2 with Delta Rule. arXiv:2412.06464.